

ADRITA AFSARA KHAN

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SUMMARY

Full-stack software developer and Computer Science graduate (Toronto Metropolitan University, 2026) who builds and ships web applications end-to-end across the JavaScript and Python stacks — React / Next.js front ends, Node / Express and Django back ends, and SQL and NoSQL databases. Hands-on experience with real-time systems (WebRTC), multi-tenant architecture, REST APIs, authentication, and cloud deployment. Seeking software engineering and full-stack developer roles.

TECHNICAL SKILLS

Languages: Java, Python, SQL, HTML/CSS, JavaScript, TypeScript, C, Bash, PHP, Assembly, Prolog, Rust, Haskell, Elixir
Frameworks & Libraries: React, Next.js, Node.js, Express, Django, LiveKit, Excalidraw, Tailwind CSS
Databases & Cloud: MySQL, PostgreSQL, MongoDB, Firebase, Cloudinary, Vercel, Render
Tools & Practices: Git/GitHub, REST APIs, WebRTC, token-based auth, Linux (Ubuntu), VS Code, Wireshark, VMware
Spoken Languages: English, French, Bengali

PROJECTS

SynqSpace — Real-Time Virtual Classroom

[Live Demo](#) · 2026

- Built a browser-based one-on-one tutoring platform with live video/audio and a shared, real-time collaborative whiteboard that lets tutor and student draw together in the same session.
- Engineered whiteboard sync over LiveKit (WebRTC) data channels using delta updates and version-based conflict reconciliation, with chunked binary transfer for sharing images and PDF pages onto the board.
- Implemented server-side, role-based access control: a token API mints per-role LiveKit JWTs (teacher gets moderation rights, student does not) so credentials never reach the browser; added a teacher-controlled draw-permission toggle and a mobile-responsive video/board layout.

Stack: Next.js (App Router), TypeScript, React, LiveKit, Excalidraw, Tailwind CSS.

Kumon Check-In / Check-Out System — Multi-Tenant Attendance App

In Progress · [Demo Video](#) · 2026

- Built a multi-tenant attendance system for two learning centres and 200+ student records, where students check in and out by scanning a barcode at a front-desk board that updates in real time.
- Added a colour-coded countdown board (on-track / 15-min warning / overdue) and an instant parent email on check-out, queued through a background worker so a scan never blocks on email delivery.
- Enforced per-tenant data isolation via a custom query manager so each owner sees only their own centres — making the system reusable by other centres/owners — and added streamed CSV export (date-range, timezone-aware) plus an append-only audit trail for a tamper-evident record.

Stack: Python, Django, MySQL, django-q2, Amazon SES, django-simple-history.

Picture Gallery — Full-Stack MERN App (Deployed)

[Repo](#) · 2026

- Built and deployed a full-stack image gallery (React, Node/Express, MongoDB) with token-based authentication and full CRUD for user photos.
- Deployed across a three-tier cloud setup — frontend on Vercel, backend on Render, database on MongoDB Atlas — wired with environment variables and CORS; images stored on Cloudinary with URLs persisted in MongoDB.

Personal Portfolio Website

adritakhan.ca · 2024–2026

- Built a live, multi-page site on a custom domain with a GitHub-API projects feed, a keyword-matching chat assistant, a PHP contact form with spam filtering, and Firebase-backed likes.

EDUCATION

Honours B.Sc., Computer Science — Toronto Metropolitan University, Toronto, ON

2022–2026

Honours B.Sc. with Distinction, Life Sciences — University of Toronto Mississauga

2016–2021

Double Major: Biology for Health Sciences & French Studies. U of T Scholar; Lieutenant Governor's Award (600+ volunteer hours).

EXPERIENCE

Primary Instructor — Kumon Math & Reading, Mississauga, ON

2016–Present

- Break down complex concepts into clear, structured steps for learners at varying levels, and track progress with data to target improvement.

Software Engineering Fellow — Headstarter AI (Remote)

2024

- Selected for an intensive, project-based fellowship in full-stack web development (React, Next.js); collaborated on weekly build sprints under deadline.

Owner — Afsara's Elegant Jewellery

2020–Present

- Run an online business end-to-end — e-commerce, digital marketing (Instagram, Facebook), product listings, customer communication, and order fulfillment.

LEADERSHIP & RESEARCH

Vice President, Boss Women Club · Facilitator, FRE272 Study Group · Mentor, Erindale Biology Society · Coordinator, UTM Helping Hands — led study sessions, planned events, and managed budgets (2017–2021). Volunteer, The Opportunity Exchange IT Career Summit (2026).

Research Student, U of T Mississauga (2019–2020) — co-conducted a study on transformative learning in science courses (survey design) and authored a cross-disciplinary literature review.